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ML Lab 2

Programmazione Funzionale
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Exercise L2.1

- Write a function `cube` that computes the cube of a real number
- For instance
  - `cube (2.9) = 24.389`

Exercise L2.2

- Write a function `min3` that computes the smallest component of a tuple of type int*int*int
- For instance
  - `min3 (2,3,4) = 2`
  - `min3 (3,2,4) = 2`

Exercise L2.3

- Write a function `third` that computes the third element of a list (it doesn't have to work properly on shorter lists).

- For instance,

```
third [2,3,4] = 4
third [2,3,4,5] = 4
```

Exercise L2.4

- Write a function `reverse` that reverses a tuple of length 3
- For instance,

```
reverse (1,2,3) = (3,2,1)
reverse (1.0,2,"a") = ("a",2,1.0)
```

Exercise L2.5

- Write a function `thirdchar` that finds the string containing the third character of a string (it doesn't have to work properly on shorter strings)
- For instance

  `thirdchar "abcd"` = `"c“`
- You can use support functions

Exercise L2.6

- Write a function `cycle` that cycles a list once, i.e., convert `[a1,...,an]` to `[a2,...,an,a1]`. It doesn't have to work on the empty list.
- For instance
  - `cycle [1,2,3,4] = [2,3,4,1]`
  - `cycle [1] = [1]`

Exercise L2.7

- Write a function `min_max_pair` that, given 3 integers (a tuple), produce a pair consisting of the smallest and the largest value among the 3 integers
- For instance
  - `min_max_pair (1,2,3) = (1,3)`
  - `min_max_pair(3,4,2) = (2,4)`
- You can use support functions

Exercise L2.8

- Write a function `sort` that, given three integers (a tuple), produce a list of them in sorted order
- For instance
  - `sort (1,2,3) = [1,2,3]`
  - `sort (1,3,2) = [1,2,3]`
- You can use support functions

Exercise L2.9

- Write a function `rnd` that rounds a real number to the nearest decimal (10^{th}) digit
- For instance
  - `rnd(2.56) = 2.6`
  - `rnd (5.678) = 5.7`
  - `rnd (3.3) = 3.3`
  - `rnd (4.128) = 4.1`

Exercise L2.10

- Write a function `rem` that, given a list, removes the second element. It doesn't need to work on lists shorter than 2.
- For instance
  - `rem [1,2,3,4] = [1,3,4]`
  - `rem [1,2] = [1]`